

# Phase 2 Documentation – Fabulari Chat System

3813ICT Full Stack Development

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GitHub: <https://github.com/HadiAlk4/Chat-App-System>

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## 1 Overview

Fabulari is a MEAN-stack chat application: MongoDB (native Node driver), Express 5, Angular 22 (standalone components, PaperCSS) and Node.js, with Socket.io for real-time chat and live request queues. Users chat in rooms that belong to groups. Three roles, Super Admin, Group Admin (GA) and User, control what each person can see and do. Phase 1 kept most data in a JSON file or in mock arrays. In Phase 2 every screen reads and writes MongoDB through the API.

**Running it:** start MongoDB on localhost:27017; cd backend && npm start (port 3000); cd frontend && npm start (port 4200).

## 2 Specifications and Requirements

Requirements come from the brief, the client meeting (Week 2) and the published clarifications. Status reflects the code on `main`.

| Area          | Requirement                                                                                                                                                                                               | Status |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Stack         | MEAN stack. MongoDB replaces <code>fakeData.json</code> . Socket.io handles real-time events.                                                                                                             | Done   |
| Auth          | Basic email and password login. Passwords are hashed with <code>bcrypt</code> . OAuth is not used.                                                                                                        | Done   |
| Password rule | At least 8 characters, alphanumeric only, at least one uppercase letter. Checked on signup and password change.                                                                                           | Done   |
| Bootstrap     | The first account registered becomes the only Super Admin. Signup shows a note when the database has no users.                                                                                            | Done   |
| Super Admin   | Handles requests only. Cannot chat, read chat history, join or propose groups. Creates a group only by approving a user proposal, and the proposer becomes the first GA. Cannot delete or ban themselves. | Done   |
| Global ban    | Approving an account deletion removes the user and all their messages, and adds their email to a permanent ban list. Banned emails cannot log in or register again.                                       | Done   |
| Audit log     | Every Super Admin decision is stored with a timestamp. A dedicated page filters by action type and date range.                                                                                            | Done   |
| Groups        | Unique group names. Any user can view all groups and request to join. No limit on how many groups a user joins or administers.                                                                            | Done   |
| Age limit     | Min age applies to every room in the group. Users under it cannot request to join. Raising it removes members who are too young, but cannot remove the last GA.                                           | Done   |
| GA settings   | Edit name, description ( $\leq 250$ chars), min age and theme without Super Admin approval. A confirm dialog lists the changes first.                                                                     | Done   |

| Area           | Requirement                                                                                                                                                               | Status   |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Rooms          | GA creates, renames and deletes rooms. Users propose rooms and the GA approves or rejects them. Zero to unlimited rooms per group.                                        | Done     |
| Members        | GA can promote members to GA. A group always keeps at least one GA. GA sees the allowed and banned member lists.                                                          | Done     |
| Group bans     | Bans are permanent and there is no un-ban. Users can ask a GA to ban someone. A ban on a GA goes to the Super Admin instead.                                              | Done     |
| Step down      | A GA cannot step down if they are the only GA or have pending Super Admin requests.                                                                                       | Done     |
| Group deletion | A GA requests deletion and the Super Admin approves it. The group and its data are removed, and the GA is demoted if they administer no other group.                      | Done     |
| Requests       | Rejections need a reason. Pending requests cannot be cancelled. Users see pending and rejected requests in Request History.                                               | Done     |
| Messaging      | Single-channel, single-thread. Text plus PNG, JPEG and GIF images up to 2MB. No external links. Shows sender name and timestamp. Users can delete their own messages.     | Done     |
| History        | Only the last 5 messages are loaded when entering a room. Live messages are not capped while the user stays. A new room starts blank.                                     | Done     |
| Real-time      | Join and leave toasts (closable with X), live participant sidebar, GA badge in the sidebar, and a toast when a join request is accepted or denied.                        | Done     |
| Cascade delete | When an account deletion is approved, that user's messages disappear from every open chat straight away.                                                                  | Done     |
| Theme          | The group's theme (light/dark) applies to its chat rooms. A user can override it with a personal theme that applies across all groups.                                    | Done     |
| Profile        | Username (changeable), email (fixed, unique ID), password change with current password, DOB/age, role, profile picture ( $\leq 2$ MB), memberships. Profiles are private. | Done     |
| Responsive     | Desktop first and usable at tablet width (PaperCSS grid).                                                                                                                 | Done     |
| HTTPS          | Messages served over HTTPS.                                                                                                                                               | Not done |

**Known limitations.** The server runs on HTTP. The API trusts the username sent by the client and has no session token, so role checks happen in the Angular guard and in selected routes. The upload filter also accepts WebP, which the spec does not list.

### 3 Server-side API

Base URL `http://localhost:3000`. Bodies are JSON (uploads use `multipart/form-data`). Most responses have the form `{ ok, message, ...data }`. Validation failures return `ok:false` with a 400, 403 or 404 status. Server errors return 500. GET list endpoints return an array.

#### 3.1 REST endpoints

| Method                          | Endpoint                            | Input                               | Behaviour                                                                                                |
|---------------------------------|-------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------|
| <i>Authentication and users</i> |                                     |                                     |                                                                                                          |
| POST                            | <code>/api/auth</code>              | email, password                     | bcrypt compare. Blocks banned emails. Returns the user without the password.                             |
| GET                             | <code>/api/signup/first-user</code> | —                                   | <code>{ firstUser }</code> . True when no users exist yet.                                               |
| POST                            | <code>/api/signup</code>            | username, email, password, dob, age | Checks the password rule, the ban list and duplicate emails. The first user becomes <b>super-admin</b> . |

| Method                                                                                     | Endpoint                               | Input                                                     | Behaviour                                                                                                                               |
|--------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| PATCH                                                                                      | /api/users/username                    | email, newUsername                                        | Must be unique. Renames the user in groups, messages and requests.                                                                      |
| PATCH                                                                                      | /api/users/password                    | email, currentPassword, newPassword                       | Checks the current password, then applies the rule and stores the new hash.                                                             |
| PATCH                                                                                      | /api/users/theme                       | email, isDarkMode, usePersonalTheme                       | Saves the personal chat theme override.                                                                                                 |
| <i>Groups, rooms and members</i>                                                           |                                        |                                                           |                                                                                                                                         |
| GET                                                                                        | /api/groups                            | –                                                         | All groups.                                                                                                                             |
| GET                                                                                        | /api/groups/user/:username             | –                                                         | Groups where the user is a member or admin.                                                                                             |
| GET                                                                                        | /api/groups/:groupName                 | –                                                         | One group, 404 if missing.                                                                                                              |
| PATCH                                                                                      | /api/groups/:groupName                 | groupName, groupDescription, minAge, themeColor, username | GA only (403 otherwise). Unique name, description ≤250 chars. Removes members under the new min age. A rename updates every collection. |
| POST                                                                                       | /api/groups/:groupName/leave           | username                                                  | Leave the group. The only GA cannot leave.                                                                                              |
| POST                                                                                       | /api/groups/:groupName/rooms/direct    | roomName                                                  | GA adds a room. Names are unique within the group, ignoring case.                                                                       |
| PATCH                                                                                      | /api/groups/:groupName/rooms/rename    | oldName, newName                                          | Rename a room.                                                                                                                          |
| DELETE                                                                                     | /api/groups/:groupName/rooms/:roomName | –                                                         | Delete a room.                                                                                                                          |
| POST                                                                                       | /api/rooms                             | groupName, roomName                                       | Older direct add from Phase 1.                                                                                                          |
| PATCH                                                                                      | /api/groups/:g/members/:u/promote      | –                                                         | Add the member to <b>admins</b> .                                                                                                       |
| POST                                                                                       | /api/groups/:g/members/:u/remove       | –                                                         | Remove the member. The only GA cannot be removed.                                                                                       |
| POST                                                                                       | /api/groups/:g/members/:u/ban          | –                                                         | Permanent group ban ( <b>bannedMembers</b> ). The only GA cannot be banned.                                                             |
| POST                                                                                       | /api/groups/:g/admins/:u/step-down     | –                                                         | Blocked if they are the only GA or have pending Super Admin requests. Demotes their role if they administer no other group.             |
| <i>Request queues (each has create, list, approve, reject; reject needs <b>reason</b>)</i> |                                        |                                                           |                                                                                                                                         |
| POST, GET                                                                                  | /api/group-requests                    | groupName, description, minAge, themeColor, creator       | Group proposal to the Super Admin. The name must be free. The Super Admin cannot propose.                                               |
| PATCH                                                                                      | /api/group-requests/:id/approve reject | performedBy, reason                                       | Approve creates the group with a <b>Main Room</b> and the proposer as GA. Both are audited.                                             |
| POST, GET                                                                                  | /api/join-requests                     | groupName, username                                       | Checks age, ban and duplicate requests. The Super Admin cannot join.                                                                    |
| PATCH                                                                                      | /api/join-requests/:id/approve reject  | reason                                                    | Approve adds the user to <b>members</b> .                                                                                               |
| POST, GET                                                                                  | /api/room-requests                     | groupName, roomName, username                             | Members only. The room name must be free.                                                                                               |
| PATCH                                                                                      | /api/room-requests/:id/approve reject  | reason                                                    | Approve adds the room.                                                                                                                  |
| POST, GET                                                                                  | /api/group-ban-requests                | groupName, targetUsername, requestedBy                    | Routed to the GA, or to the Super Admin if the target is a GA. Filter with <b>destination</b> .                                         |

| Method                         | Endpoint                                          | Input                             | Behaviour                                                                                |
|--------------------------------|---------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------|
| PATCH                          | /api/group-ban-requests/:id/approve reject        | reason                            | Approve applies the permanent group ban.                                                 |
| POST, GET                      | /api/group-deletion-requests                      | groupName, requestedBy, reason    | GA only. One pending request per group.                                                  |
| PATCH                          | /api/group-deletion-requests/:id/approve reject   | performedBy, reason               | Approve deletes the group, its messages and its requests, and demotes the GA. Audited.   |
| POST, GET                      | /api/account-deletion-requests                    | username                          | The user asks to delete their account. Not allowed for the Super Admin.                  |
| PATCH                          | /api/account-deletion-requests/:id/approve reject | performedBy, reason               | Approve deletes the user and their messages and bans the email. Audited as a global ban. |
| GET                            | /api/banned-emails                                | –                                 | Permanently banned emails, newest first.                                                 |
| <i>Chat, uploads and audit</i> |                                                   |                                   |                                                                                          |
| GET                            | /api/messages/:groupName/:roomName                | query: username                   | Last 5 messages, oldest first. Returns 403 for the Super Admin.                          |
| DELETE                         | /api/messages/:id                                 | –                                 | Delete a message.                                                                        |
| POST                           | /api/upload/avatar                                | form: image, username             | Image $\leq$ 2 MB. Sets <b>profilePictureUrl</b> .                                       |
| POST                           | /api/upload/chat                                  | form: image                       | Image $\leq$ 2 MB. Returns <b>fileUrl</b> (served from /uploads).                        |
| GET                            | /api/audit-logs                                   | query: action, startDate, endDate | Filtered audit log, newest first.                                                        |

## 3.2 Socket.io events

Room channels are named `groupName:roomName`. The server ignores chat events from unknown users and from the Super Admin, and drops messages that contain `http(s)://` or `www..`

| Direction     | Event                              | Payload and effect                                                                                                                                                                                                                                                                              |
|---------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| client→server | join-room, leave-room              | {groupName, roomName, username}. Joins or leaves the channel and updates who is present.                                                                                                                                                                                                        |
| client→server | send-message                       | {groupName, roomName, senderUserName, content, imageUrl}. Saves to <code>messages</code> , then broadcasts it.                                                                                                                                                                                  |
| client→server | delete-message                     | {groupName, roomName, messageId}. Deletes from MongoDB, then broadcasts.                                                                                                                                                                                                                        |
| server→room   | new-message, message-deleted       | The saved message, or {messageId}.                                                                                                                                                                                                                                                              |
| server→room   | user-joined, user-left, room-users | Trigger toasts and the live participant list.                                                                                                                                                                                                                                                   |
| server→all    | <queue>-created, <queue>-resolved  | For group, join, room, group-ban, group-deletion and account-deletion requests. Queues and Request History refresh live. <code>join-request-resolved</code> triggers the accept/deny toast. <code>account-deletion-request-resolved</code> removes the deleted user's messages from open chats. |

## 4 Angular Architecture

Standalone components with routing in `app.routes.ts`. The logged-in user is kept in `sessionStorage`, so each browser tab has its own login. `authGuard` protects every route except login and signup. It checks `expectedRole` for Super Admin pages and sends a Super Admin away from `chat`, `my-memberships` and `group-settings`.

### 4.1 Components

| Component           | Route                  | Responsibility                                                                                                                                                                                                                                                   |
|---------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Login               | /login                 | Posts credentials to <code>/api/auth</code> , stores the user, and routes by role (Super Admin dashboard or dashboard). Shows API errors, including the banned message.                                                                                          |
| Signup              | /signup                | Registration with age from DOB and an optional profile doodle. Shows a note when the next account will be the Super Admin.                                                                                                                                       |
| Dashboard           | /dashboard             | Lists and searches groups, proposes a group (modal), and requests to join. Groups the user is already in are hidden. Both actions are hidden for the Super Admin.                                                                                                |
| MyMemberships       | /my-memberships        | The user's groups, filtered by name and role. Opens chat, settings, or leave.                                                                                                                                                                                    |
| GroupSettings       | /group-settings        | GA-only tabs. <i>General</i> : edit and save with a confirm dialog. <i>Rooms</i> : add, rename, delete. <i>Members</i> : promote, ban, banned list. <i>Requests</i> : join, room and ban queues. Also step down and request deletion. Updates live over sockets. |
| RequestHistory      | /request-history       | The user's pending and rejected requests with reasons. Tabs to propose a room and to request a ban.                                                                                                                                                              |
| Chat                | /chat                  | Room list, last-5 history plus live messages, image attachments, delete own message, toasts, participant sidebar with GA badge, and the group or personal theme.                                                                                                 |
| UserProfileSettings | /user-profile-settings | Change username, password and theme. Upload a profile picture. Request account deletion.                                                                                                                                                                         |
| SuperAdminDashboard | /super-admin-dashboard | Live queues for group proposals, GA ban requests, group deletions and account deletions, plus the banned-email list. Every rejection needs a reason.                                                                                                             |
| SuperAdminAuditLog  | /super-admin-audit-log | Audit table filtered by action type and date range.                                                                                                                                                                                                              |

## 4.2 Services

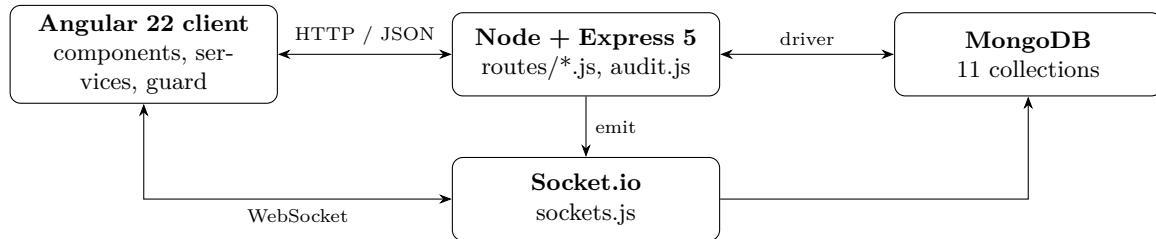
- **AuthService**: get, set and clear the session user; logout.
- **GroupService**: all group, room and member endpoints, plus every request queue (group, join, room, ban, deletion).
- **ChatService**: room history over HTTP. Emits and listens to chat socket events (`join-room`, `send-message`, `new-message`, `room-users`, and so on).
- **SocketService**: typed observables for every `*-created` and `*-resolved` request event.
- **AccountService**: account deletion requests, banned emails, and username, password and theme updates.
- **UploadService**: avatar and chat image uploads (`FormData`).
- **AuditService**: audit log query with filters.

## 4.3 Models (TypeScript interfaces)

`Group`, `GroupRequest`, `JoinRequest`, `RoomRequest`, `GroupBanRequest`, `GroupDeletionRequest`, `AccountDeletionRequest`, `BannedEmail`, `AuditLog`, `ChatMessage`. They mirror the MongoDB documents in Section 5.2. Request models share a `status` of `pending`, `approved` or `rejected`, plus `createdAt`, `reviewedAt` and `rejectionReason`.

## 5 Design Documents

### 5.1 System architecture



`app.js` builds the Express app (`createApp(io)`). `index.js` attaches the HTTP server and Socket.io. Because of this split, tests can import the app without starting a server, using a stub `io`. Business rules are pulled out into small, pure helpers in `backend/lib/` (password policy, age check, link check, last-5 window).

### 5.2 Data structures (MongoDB collections)

| Collection                           | Fields                                                                                                                                                               |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>users</code>                   | username, email (unique ID), password (bcrypt), dob, age, role ( <code>super-admin group-admin user</code> ), valid, isDarkMode, usePersonalTheme, profilePictureUrl |
| <code>groups</code>                  | groupName (unique), groupDescription, minAge, themeColor ( <code>light dark</code> ), admins[], members[], rooms[], bannedMembers[]                                  |
| <code>messages</code>                | groupName, roomName, senderUserName, content, imageUrl, timestamp                                                                                                    |
| <code>groupRequests</code>           | groupName, groupDescription, minAge, themeColor, creatorUserName, creatorEmail + request fields                                                                      |
| <code>joinRequests</code>            | groupName, username + request fields                                                                                                                                 |
| <code>roomRequests</code>            | groupName, roomName, username + request fields                                                                                                                       |
| <code>groupBanRequests</code>        | groupName, targetUsername, targetEmail, targetRole, requestedBy, requestedByRole, destination ( <code>group-admin super-admin</code> ) + request fields              |
| <code>groupDeletionRequests</code>   | groupName, requestedBy, requestedByRole, reason + request fields                                                                                                     |
| <code>accountDeletionRequests</code> | username, email, role + request fields                                                                                                                               |
| <code>bannedEmails</code>            | email, username, bannedAt, reason                                                                                                                                    |
| <code>auditLogs</code>               | timeStamp, actionPerformed, target, performedBy (insert only)                                                                                                        |

*Request fields* are status, createdAt, reviewedAt and rejectionReason. Groups store room names and usernames as arrays instead of references, so a single read loads a whole group. When a user or group is renamed, the server updates the name in every collection.

### 5.3 Storyboards

The Phase 1 storyboards (desktop and tablet) still describe the layout, so they were kept. They are in Appendix A and in `assignment_material/Fabulari Design/`. Changes made in Phase 2:

- The Group Settings *General* tab is a real form and asks for confirmation before saving.
- Request History has a *Request Ban* tab.
- The Super Admin dashboard has live GA ban and group deletion queues.
- Chat has image attachments and a delete button on your own messages.
- Every Super Admin rejection asks for a reason.

## 6 Testing

### 6.1 Tools and methodology

- **Backend unit tests**, Mocha with Node `assert` (`npm run unitTest`). Test the pure helpers in `backend/lib`.

- **Backend integration tests**, Mocha, Chai and chai-http (`npm test`). Each test calls `createApp()` against a separate `chat-app-test` database. The database is wiped before and after every test, so no test depends on another. Each endpoint has at least one success case and one failure case.
- **Angular unit tests**, Vitest with TestBed via `ng test`. Services are replaced with test doubles, so components are tested without a backend.
- **End-to-end tests**, Cypress (`npm run cypress:run`), against the real frontend and a backend started with `MONGO_DB_NAME=chat-app-test`. A `resetDb` task drops the test database before each spec.

Before any tests were written, the logic was moved into testable helpers and the app setup was split from `listen()`. The end-to-end tests found four screens that did not refresh after data loaded (login errors, dashboard, memberships, chat), and these were fixed.

**Results (run 1 Oct 2026):** all 150 automated tests pass: backend unit 14/14, backend integration 100/100, Angular 27/27 and Cypress 9/9. The full output of each run is in Section 6.3.

## 6.2 Automated tests

| Suite       | Target                                            | #  | Cases                                                                                                                                                      |
|-------------|---------------------------------------------------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unit        | <code>isValidPassword</code>                      | 4  | accepts valid; rejects too short, no uppercase, symbol                                                                                                     |
| Unit        | <code>isUnderMinAge</code>                        | 3  | under min; equal to min; missing age not treated as under                                                                                                  |
| Unit        | <code>containsExternalLink</code>                 | 3  | detects http, www; allows plain text                                                                                                                       |
| Unit        | <code>latestMessagesOldestFirst</code>            | 4  | keeps latest 5; exactly 5; fewer than 5; oldest-first order                                                                                                |
| Integration | <code>/api/auth</code> , <code>/api/signup</code> | 6  | valid login; wrong password; banned login; first user is Super Admin; weak password; banned email signup                                                   |
| Integration | <code>/api/users/*</code>                         | 6  | rename; missing name; change password; wrong current password; save theme; missing theme flags                                                             |
| Integration | <code>/api/groups*</code>                         | 11 | list all / empty; user groups / none; get one / 404; leave / missing username; save settings and remove under-age members; missing description; non-GA 403 |
| Integration | rooms                                             | 8  | add / missing name; rename / missing new name; delete / 404; <code>POST /api/rooms</code> add / unknown group                                              |
| Integration | members                                           | 8  | promote / 404; remove / only GA; ban / 404; step down / only GA                                                                                            |
| Integration | group requests                                    | 8  | create / missing fields; list / empty filter; approve creates group with GA / not pending; reject with reason / no reason                                  |
| Integration | join requests                                     | 8  | create / under age; list / empty filter; approve adds member / not pending; reject with reason / no reason                                                 |
| Integration | room requests                                     | 8  | create / missing fields; list / empty filter; approve adds room / not pending; reject with reason / no reason                                              |
| Integration | group ban requests                                | 8  | create / missing fields; list by destination / empty; approve bans / 404; reject with reason / no reason                                                   |
| Integration | group deletion                                    | 8  | create / missing reason; list / empty; approve deletes and demotes / 404; reject with reason / no reason                                                   |
| Integration | account deletion, banned emails                   | 10 | create / Super Admin blocked; list / empty; approve deletes and bans / 404; reject with reason / no reason; banned list / empty                            |
| Integration | <code>/api/messages</code>                        | 5  | last 5 oldest first; missing username 400; Super Admin 403; delete / 404                                                                                   |
| Integration | <code>/api/upload/*</code>                        | 4  | avatar PNG / no file; chat PNG / no file                                                                                                                   |
| Integration | <code>/api/audit-logs</code>                      | 2  | returns logs; filters by action                                                                                                                            |
| Angular     | App                                               | 2  | creates root; renders router outlet                                                                                                                        |
| Angular     | authGuard                                         | 3  | logged out to login; user blocked from Super Admin page; Super Admin blocked from chat                                                                     |
| Angular     | Login                                             | 3  | empty fields no HTTP; user to dashboard; Super Admin to Super Admin dashboard                                                                              |
| Angular     | Signup                                            | 4  | age for past, empty and future DOB; empty form not posted                                                                                                  |
| Angular     | Dashboard                                         | 2  | proposal sent via service; joined groups hidden                                                                                                            |
| Angular     | Chat                                              | 3  | link not sent; plain text sent; history shows sender name                                                                                                  |

| Suite   | Target                           | # | Cases                                                                                             |
|---------|----------------------------------|---|---------------------------------------------------------------------------------------------------|
| Angular | GroupSettings                    | 4 | form filled from group; confirm before raising min age; rejection needs reason; non-GA redirected |
| Angular | RequestHistory,<br>MyMemberships | 2 | renders a join request; renders a membership                                                      |
| Angular | SuperAdminDashboard              | 1 | renders pending proposals                                                                         |
| Angular | SuperAdminAuditLog               | 2 | renders logs; applies action and date filters                                                     |
| Angular | UserProfileSettings              | 1 | shows error on wrong current password                                                             |
| E2E     | login.cy.ts                      | 3 | logs in and fills dashboard; sends credentials to API; error on empty submit                      |
| E2E     | access.cy.ts                     | 3 | logged-out redirect; logged-in reaches dashboard; bad credentials stay on login                   |
| E2E     | admin-group.cy.ts                | 1 | after Super Admin approval the group leaves the creator's join list and appears in My Memberships |
| E2E     | chat.cy.ts                       | 2 | two users: other member's message shows their name; link message not shown                        |

## 6.3 Test run output

Output from the four test commands, pasted as printed. Colour codes, Node/npm warnings and a Cypress start-up warning about clearing old screenshots are removed, and tick, cross and box-drawing characters are replaced with ASCII so the PDF font can display them.

### Backend unit tests (cd backend && npm run unitTest)

```
> chat-app-system@1.0.0 unitTest
> mocha unitTest

isUnderMinAge
  under the minimum
    ok returns true when the age is below the group minimum
  equal to the minimum
    ok returns false when the age matches the group minimum
  missing age
    ok does not treat a missing age as under the minimum

containsExternalLink
  http link
    ok detects an http URL
  www link
    ok detects a www host
  plain text
    ok allows text that is not a link

latestMessagesOldestFirst
  more than five
    ok keeps only the latest five messages
  exactly five
    ok returns all five messages
  fewer than five
    ok returns every message when the room has fewer than five
  oldest-first order
    ok returns the window with the oldest message first

isValidPassword
  valid password
    ok accepts 8 alphanumeric characters with an uppercase letter
  too short
    ok rejects a password shorter than 8 characters
  missing uppercase
    ok rejects a long password with no uppercase letter
  symbol rejected
    ok rejects a password that contains a symbol

14 passing (6ms)
```

### Backend integration tests (cd backend && npm test)

```
> chat-app-system@1.0.0 test
> MONGO_DB_NAME=chat-app-test mocha --require ./integrationTest/hooks.js --timeout 20000 "integrationTest/*.test.js"
```

```

account deletion routes
  POST /api/account-deletion-requests
    ok queues a deletion request (46ms)
    ok rejects a super admin deletion request
  GET /api/account-deletion-requests
    ok returns pending requests
    ok returns an empty list when the status filter matches nothing
  PATCH /api/account-deletion-requests/:id/approve
    ok deletes the user and bans the email
    ok returns 404 when the request is not pending
  PATCH /api/account-deletion-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason
  GET /api/banned-emails
    ok returns banned emails
    ok returns an empty list when nobody is banned

audit routes
  GET /api/audit-logs
    ok returns stored audit logs
    ok filters logs by action type

auth routes
  POST /api/auth
    ok logs in with a valid email and password
    ok rejects a wrong password
    ok tells a permanently banned account it cannot log in
  POST /api/signup
    ok creates the first account as super admin (102ms)
    ok rejects a password without an uppercase letter
    ok rejects a permanently banned email

chat routes
  GET /api/messages/:groupName/:roomName
    ok returns the latest five messages oldest first
    ok rejects a history request with no username
    ok forbids a super admin from reading chat history
  DELETE /api/messages/:id
    ok deletes a stored message
    ok returns 404 when the message does not exist

group ban request routes
  POST /api/group-ban-requests
    ok queues a ban request against another member
    ok rejects a request that is missing fields
  GET /api/group-ban-requests
    ok returns pending requests for a destination
    ok returns an empty list when the destination filter matches nothing
  PATCH /api/group-ban-requests/:id/approve
    ok bans the target member (41ms)
    ok returns 404 when the request is not pending
  PATCH /api/group-ban-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group deletion routes
  POST /api/group-deletion-requests
    ok queues a deletion request from a group admin
    ok rejects a request that is missing a reason
  GET /api/group-deletion-requests
    ok returns pending requests
    ok returns an empty list when the status filter matches nothing
  PATCH /api/group-deletion-requests/:id/approve
    ok deletes the group and demotes the requesting admin
    ok returns 404 when the request is not pending
  PATCH /api/group-deletion-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group request routes
  POST /api/group-requests
    ok queues a proposal
    ok rejects a proposal that is missing fields
  GET /api/group-requests
    ok returns pending proposals
    ok returns an empty list when the status filter matches nothing
  PATCH /api/group-requests/:id/approve
    ok creates the group and assigns the proposer as admin
    ok rejects an id that is not a pending proposal
  PATCH /api/group-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

group routes
  GET /api/groups
    ok returns every group
    ok returns an empty list when no groups exist
  GET /api/groups/user/:username
    ok returns groups the user belongs to
    ok returns an empty list for a user who belongs to no group
  GET /api/groups/:groupName

```

```

    ok returns one group
    ok returns 404 when the group does not exist
POST /api/groups/:groupName/leave
    ok removes a member who is not the sole admin
    ok rejects a leave request with no username
PATCH /api/groups/:groupName
    ok saves settings and removes members under the new minimum age (39ms)
    ok rejects a description that is missing
    ok rejects a member who is not a Group Admin

join request routes
POST /api/join-requests
    ok creates a pending join request
    ok rejects a user who is under the group minimum age
GET /api/join-requests
    ok returns pending requests for a group
    ok returns an empty list when the status filter matches nothing
PATCH /api/join-requests/:id/approve
    ok adds the requester to the group
    ok rejects an id that is not a pending request
PATCH /api/join-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

member routes
PATCH /api/groups/:groupName/members/:username/promote
    ok adds a member to the admin list
    ok returns 404 when the user is not in the group
POST /api/groups/:groupName/members/:username/remove
    ok removes a member who is not the sole admin
    ok rejects removal of the sole administrator
POST /api/groups/:groupName/members/:username/ban
    ok permanently bans a member
    ok returns 404 when the user is not in the group
POST /api/groups/:groupName/admins/:username/step-down
    ok steps down when another group admin remains
    ok rejects a step-down by the sole group admin

room request routes
POST /api/room-requests
    ok creates a pending room proposal
    ok rejects a proposal that is missing fields
GET /api/room-requests
    ok returns pending proposals for a group
    ok returns an empty list when the status filter matches nothing
PATCH /api/room-requests/:id/approve
    ok adds the proposed room
    ok rejects an id that is not a pending proposal
PATCH /api/room-requests/:id/reject
    ok stores the rejection reason
    ok rejects a decision that has no reason

room routes
POST /api/groups/:groupName/rooms/direct
    ok adds a room
    ok rejects a missing room name
PATCH /api/groups/:groupName/rooms/rename
    ok renames a room
    ok rejects a rename that omits the new name
DELETE /api/groups/:groupName/rooms/:roomName
    ok deletes a room
    ok returns 404 when the room does not exist
POST /api/rooms
    ok adds a room to an existing group
    ok rejects a room for a group that does not exist

upload routes
POST /api/upload/avatar
    ok stores a small PNG as the profile picture
    ok rejects a request with no file
POST /api/upload/chat
    ok accepts a small PNG
    ok rejects a request with no file

user routes
PATCH /api/users/username
    ok renames the account
    ok rejects a missing new username
PATCH /api/users/password
    ok changes the password when the current password matches (166ms)
    ok rejects a wrong current password
PATCH /api/users/theme
    ok saves the chat theme flags
    ok rejects a theme update that omits the boolean flags

Database connection closed

100 passing (2s)

```

## Angular unit tests (cd frontend && ng test --watch=false --reporters=verbose)

```
> Building...
ok Building...
Initial chunk files | Names | Raw size
init-testbed.js | init-testbed | 335.90 kB |
spec-app-group-settings-group-settings.js | spec-app-group-settings-group-settings | 87.05 kB |
spec-app-request-history-request-history.js | spec-app-request-history-request-history | 72.27 kB |
styles.css | styles | 66.10 kB |
spec-app-chat-chat.js | spec-app-chat-chat | 57.30 kB |
spec-app-super-admin-dashboard-super-admin-dashboard.js | spec-app-super-admin-dashboard-super-admin-dashboard | 51.99 kB |
spec-app-dashboard-dashboard.js | spec-app-dashboard-dashboard | 33.15 kB |
spec-app-user-profile-settings-user-profile-settings.js | spec-app-user-profile-settings-user-profile-settings | 31.10 kB |
spec-app-my-memberships-my-memberships.js | spec-app-my-memberships-my-memberships | 24.00 kB |
spec-app-signup-signup.js | spec-app-signup-signup | 19.75 kB |
spec-app-super-admin-audit-log-super-admin-audit-log.js | spec-app-super-admin-audit-log-super-admin-audit-log | 18.54 kB |
spec-app-login-login.js | spec-app-login-login | 12.28 kB |
spec-app-guards-auth-guard.js | spec-app-guards-auth-guard | 4.16 kB |
spec-app-app.js | spec-app-app | 1.93 kB |
vitest-mock-patch.js | vitest-mock-patch | 988 bytes |
setup-test-setup.js | setup-test-setup | 777 bytes |

| Initial total | 817.28 kB

Application bundle generation complete. [1.170 seconds] - 2026-09-30T23:30:21.570Z

RUN v4.1.11 /Users/abdahdialkhub/Desktop/ChatApp System/Chat-App-System/frontend

stderr | src/app/login/login.spec.ts
NG0912: Component ID generation collision detected. Components '_Login' and '_Login' with selector 'app-login' generated the
↪ same component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to
↪ force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/super-admin-dashboard/super-admin-dashboard.spec.ts
NG0912: Component ID generation collision detected. Components '_SuperAdminDashboard' and '_SuperAdminDashboard' with selector
↪ 'app-super-admin-dashboard' generated the same component ID. To fix this, you can change the selector of one of those
↪ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/super-admin-audit-log/super-admin-audit-log.spec.ts
NG0912: Component ID generation collision detected. Components '_SuperAdminAuditLog' and '_SuperAdminAuditLog' with selector
↪ 'app-super-admin-audit-log' generated the same component ID. To fix this, you can change the selector of one of those
↪ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/group-settings/group-settings.spec.ts
NG0912: Component ID generation collision detected. Components '_GroupSettings' and '_GroupSettings' with selector
↪ 'app-group-settings' generated the same component ID. To fix this, you can change the selector of one of those components or
↪ add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/chat/chat.spec.ts
NG0912: Component ID generation collision detected. Components '_Chat' and '_Chat' with selector 'app-chat' generated the same
↪ component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to force a
↪ different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/request-history/request-history.spec.ts
NG0912: Component ID generation collision detected. Components '_RequestHistory' and '_RequestHistory' with selector
↪ 'app-request-history' generated the same component ID. To fix this, you can change the selector of one of those components
↪ or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

stderr | src/app/dashboard/dashboard.spec.ts
NG0912: Component ID generation collision detected. Components '_Dashboard' and '_Dashboard' with selector 'app-dashboard'
↪ generated the same component ID. To fix this, you can change the selector of one of those components or add an extra host
↪ attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/login/login.spec.ts > Login > shows an error and does not call HTTP when the fields are empty 95ms
ok |frontend| src/app/super-admin-dashboard/super-admin-dashboard.spec.ts > SuperAdminDashboard > renders pending group
↪ proposals from the group service 111ms
ok |frontend| src/app/login/login.spec.ts > Login > navigates a regular user to the dashboard after a successful login 7ms
ok |frontend| src/app/login/login.spec.ts > Login > navigates a super admin to the super admin dashboard 10ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > fills the form from the loaded group 93ms
ok |frontend| src/app/super-admin-audit-log/super-admin-audit-log.spec.ts > SuperAdminAuditLog > renders the mocked audit log
↪ 60ms
ok |frontend| src/app/super-admin-audit-log/super-admin-audit-log.spec.ts > SuperAdminAuditLog > applies the action and date
↪ filters to the mocked list 61ms
stderr | src/app/user-profile-settings/user-profile-settings.spec.ts
NG0912: Component ID generation collision detected. Components '_UserProfileSettings' and '_UserProfileSettings' with selector
↪ 'app-user-profile-settings' generated the same component ID. To fix this, you can change the selector of one of those
↪ components or add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a logged-out visitor to the login page 3ms
ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a user away from a super admin page 1ms
ok |frontend| src/app/guards/auth-guard.spec.ts > authGuard > sends a super admin away from chat 1ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > asks for confirmation before saving a higher
↪ minimum age 24ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings > does not call the API when a join rejection has
↪ no reason 15ms
ok |frontend| src/app/group-settings/group-settings.spec.ts > GroupSettings access > sends a regular member away from group
↪ settings 12ms
stderr | src/app/my-memberships/my-memberships.spec.ts
```

```

NG0912: Component ID generation collision detected. Components '_MyMemberships' and '_MyMemberships' with selector
↪ 'app-my-memberships' generated the same component ID. To fix this, you can change the selector of one of those components or
↪ add an extra host attribute to force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/request-history/request-history.spec.ts > RequestHistory > renders the mocked join request 128ms
ok |frontend| src/app/chat/chat.spec.ts > Chat > does not send a message that contains an external link 140ms
ok |frontend| src/app/chat/chat.spec.ts > Chat > sends a plain text message 22ms
ok |frontend| src/app/chat/chat.spec.ts > Chat > renders the room history with the sender display name 10ms
ok |frontend| src/app/dashboard/dashboard.spec.ts > Dashboard > sends a named proposal through the group service 120ms
ok |frontend| src/app/dashboard/dashboard.spec.ts > Dashboard > hides groups the user already belongs to from the join list
↪ 11ms
ok |frontend| src/app/user-profile-settings/user-profile-settings.spec.ts > UserProfileSettings > shows the service error when
↪ the current password is wrong 48ms
stderr | src/app/signup/signup.spec.ts
NG0912: Component ID generation collision detected. Components '_Signup' and '_Signup' with selector 'app-signup' generated the
↪ same component ID. To fix this, you can change the selector of one of those components or add an extra host attribute to
↪ force a different ID. Find more at https://v22.angular.dev/errors/NG0912

ok |frontend| src/app/my-memberships/my-memberships.spec.ts > MyMemberships > renders the mocked membership 27ms
ok |frontend| src/app/app.spec.ts > App > creates the root component 3ms
ok |frontend| src/app/app.spec.ts > App > renders the router outlet 3ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > calculates a positive age for a past date of birth 30ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > returns -1 when the date of birth is empty 8ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > returns a negative age for a future date of birth 8ms
ok |frontend| src/app/signup/signup.spec.ts > Signup > does not post when the form is empty 7ms

Test Files 12 passed (12)
Tests 27 passed (27)
Start at 09:30:21
Duration 1.64s (transform 1.38s, setup 4.26s, import 619ms, tests 1.07s, environment 6.98s)

```

## End-to-end tests (cd frontend && npx cypress run)

```

=====

(Run Starting)

+-----+
| Cypress:      16.1.0                               |
| Browser:      Electron 146 (headless) (deprecated) |
| Node Version: v26.8.2 (/opt/homebrew/Cellar/node/26.8.2/bin/node) |
| Specs:         4 found (access.cy.ts, admin-group.cy.ts, chat.cy.ts, login.cy.ts) |
| Searched:     cypress/e2e/**/*.cy.{js,jsx,ts,tsx} |
+-----+

Warning: The Electron browser is deprecated as a test browser and will be removed in a future version of Cypress.

Switch to Chrome or another installed browser to avoid a breaking change when you upgrade.

Read more about supported browsers: https://on.cypress.io/launching-browsers

-----

Running:  access.cy.ts                                     (1 of 4)

Access control
  ok redirects a logged-out visit to the dashboard back to login (2875ms)
  ok lets a logged-in user reach the dashboard (846ms)
  ok keeps a user with bad credentials on the login page (634ms)

3 passing (4s)

(Results)

+-----+
| Tests:        3                                     |
| Passing:      3                                     |
| Failing:      0                                     |
| Pending:      0                                     |
| Skipped:      0                                     |
| Screenshots:  0                                     |
| Video:        false                                 |
| Duration:     4 seconds                             |
| Spec Ran:     access.cy.ts                         |
+-----+

-----

Running:  admin-group.cy.ts                                (2 of 4)

Group approval
  ok shows a group to its creator once the Super Admin approves it (3064ms)

1 passing (3s)

(Results)

+-----+

```

```
| Tests:      1
| Passing:    1
| Failing:    0
| Pending:    0
| Skipped:    0
| Screenshots: 0
| Video:      false
| Duration:   3 seconds
| Spec Ran:   admin-group.cy.ts
+-----+

Running: chat.cy.ts (3 of 4)

Chat
  ok shows another member's message with their display name (1494ms)
  ok does not show a message that contains an external link (1042ms)

2 passing (3s)

(Results)
+-----+
| Tests:      2
| Passing:    2
| Failing:    0
| Pending:    0
| Skipped:    0
| Screenshots: 0
| Video:      false
| Duration:   2 seconds
| Spec Ran:   chat.cy.ts
+-----+

Running: login.cy.ts (4 of 4)

Login
  ok logs in and fills the dashboard (1083ms)
  ok sends the credentials to the auth API (635ms)
  ok shows an error when submitted empty (320ms)

3 passing (2s)

(Results)
+-----+
| Tests:      3
| Passing:    3
| Failing:    0
| Pending:    0
| Skipped:    0
| Screenshots: 0
| Video:      false
| Duration:   2 seconds
| Spec Ran:   login.cy.ts
+-----+

=====

(Run Finished)

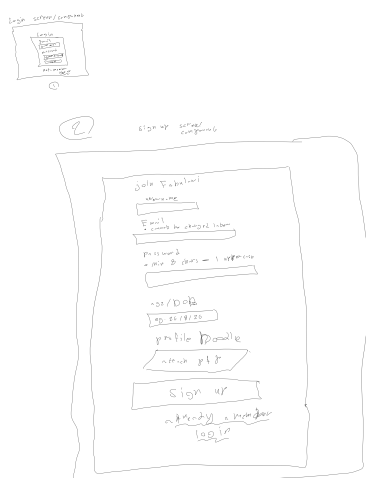
  Spec                                Tests  Passing  Failing  Pending  Skipped
+-----+
| ok  access.cy.ts                    00:04   3        3        -        -        - |
+-----+
| ok  admin-group.cy.ts               00:03   1        1        -        -        - |
+-----+
| ok  chat.cy.ts                      00:02   2        2        -        -        - |
+-----+
| ok  login.cy.ts                     00:02   3        3        -        -        - |
+-----+
| ok  All specs passed!                00:12   9        9        -        -        - |
+-----+
```

## 7 Git Workflow

Every feature was built on its own branch and merged into `main` through a GitHub pull request (221 commits, PRs #1 to #52). Each PR description lists what the branch changed. Phase 2 branches, in order:

| Branch (PR)                              | Work                                                                                           |
|------------------------------------------|------------------------------------------------------------------------------------------------|
| feature/auth-route-guard (#28)           | AuthService, route guard with roles, logout on every screen                                    |
| feature/auth-and-live-groups (#30, #31)  | bcrypt and Super Admin bootstrap, Group models and service, dashboard loads live groups        |
| feat/group-proposal-flow (#32, #33)      | Group proposal to Super Admin queue; first Socket.io events                                    |
| feature/group-join-requests (#34, #35)   | Join request model, routes, sockets, GA queue, Request History                                 |
| refactor/backend-mongodb-structure (#36) | Split backend into <code>db.js</code> and <code>routes/*</code> modules                        |
| feat/room-proposal (#37, #38)            | Room proposals wired to Group Settings                                                         |
| feature/my-memberships (#39)             | Memberships from MongoDB                                                                       |
| feature/group-settings (#40)             | Rooms and Members tabs on live data                                                            |
| feat/chat (#41, #42)                     | Messages collection, chat sockets, last-5 history, delete, live participants                   |
| feature/file-attachments (#43)           | Multer uploads for avatars and chat images                                                     |
| account-deletion-bans (#44)              | Account deletion queue, permanent email bans                                                   |
| audit-log-mongodb (#45)                  | Audit log collection and filtered page                                                         |
| user-profile-mdb (#46)                   | Username, password and theme updates                                                           |
| group-admin-remaining (#47)              | Unique names, settings save, age eviction, group bans, ban requests, step down, group deletion |
| feat/sa-request-only-scope (#48)         | Super Admin limited to requests (UI, guard, API, sockets)                                      |
| feat/chat-toast-cascade-theme (#49)      | Join toasts, live cascade delete, link blocking, personal theme                                |
| testing (#50)                            | Testable refactor, then Mocha, Chai, Vitest and Cypress suites                                 |
| cleanup-bugs (#51, #52)                  | GA check on settings, confirm dialog, first-user note, per-tab sessions                        |

## A Storyboards



Login / Sign up



Dashboard

Current user: — | Role: —

Search by Spring Name: [All Roles] Filter

| Title      | Rule  | Description     |
|------------|-------|-----------------|
| View Rules | Learn | Spring Training |

| Title      | Rule  | Description |
|------------|-------|-------------|
| View Rules | Learn |             |

| Title      | Rule  | Description     |
|------------|-------|-----------------|
| View Rules | Learn | Spring Training |

| Title      | Rule  | Description     |
|------------|-------|-----------------|
| View Rules | Learn | Spring Training |

| Title      | Rule  | Description |
|------------|-------|-------------|
| View Rules | Learn |             |

| Title      | Rule  | Description     |
|------------|-------|-----------------|
| View Rules | Learn | Spring Training |

| Title      | Rule  | Description |
|------------|-------|-------------|
| View Rules | Learn |             |

Group Name  
a

Friday Lab 10:00 AM

Link

FSD LABS

#

#

#

Group Members:

Admin:

User:

User:

Admin:

User:

Online Room Members

Join / Leave Groups notification

to be added

Send

The image contains two hand-drawn sketches of user interface elements.

The top sketch is titled "Account Settings" and includes a "User name:" label. It features a "Darkboard" button, a "Identify & Preferences" section with a "Name" input field and a "Save" button, a "Dark Mode Theme" section with a "Dark" button and a list item "Dark for the dark screen.", and a "Request Account Deletion..." button.

The bottom sketch is titled "Profile Doodle" and shows a "Profile Doodle" input field with a smiley face icon. Below it is a "Want to change your look?" section with a "Create New Doodle Below" button. At the bottom is a "Create New Doodle" button.

**S**

Super Admin Audit Log

Name: \_\_\_\_\_ to \_\_\_\_\_ All Actions ☐ Apply

Role: \_\_\_\_\_

| Super Admin Dashboard | #   | Timestamp | Action performed | Target ID |
|-----------------------|-----|-----------|------------------|-----------|
| log out               | 1 - | _____     | _____            | _____     |
|                       | 2 - | _____     | _____            | _____     |
|                       | 3 - | _____     | _____            | _____     |
|                       | 4 - | _____     | _____            | _____     |
|                       | 5 - | _____     | _____            | _____     |

15